Yang Wu

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APPOINTMENTS

- **AI Research Scientist II**, Axon, Boston, MA, *Apr 2026 – Present*
- **Research Assistant / Teaching Assistant**, Worcester Polytechnic Institute, Worcester, MA, *Aug 2021 – Mar 2026*
- **2025 Intern - Research Scientist/ Engineer**, Adobe, San Jose, CA, *May 2025 – Aug 2025*
- **Software Engineer Intern (AML-Enterprise) - 2024 Summer (PhD)**, ByteDance, San Jose, CA, *May 2024 – Aug 2024*
- **Bioinformatics Intern**, EMD Serono Research Center, Billerica, MA, *Jan 2020 – Jun 2020*

EDUCATION

Worcester Polytechnic Institute **Worcester, MA**
Ph.D. in Computer Science *May 2026*
Dissertation title: - Advancing the Proactive Capacity of LLMs in Cost-Sensitive Domains
Advisor: Xiaozhong Liu

Northeastern University **Boston, MA**
MPS in Analytics *Apr 2021*

Wuhan University **Wuhan, China**
B.S. in Material Engineering *Jun 2018*

RESEARCH HIGHLIGHT

- Research focuses on trustworthy NLP and large language models, combining data-efficient training, human-centered feedback, and domain-adaptive reasoning, applied to legal reasoning, biomedical discovery, personalized education, and public safety.
- Lead AI Research projects on data-efficient LLM training, proactive LLM assistants, multimodal LLM systems, and generative AI applications.
- Published 16 peer-reviewed articles at top-tier AI/NLP venues (ACL, EMNLP, NAACL, CIKM, WWW), with 4 additional papers currently under review and 3 papers in preparation.

PEER REVIEWED CONFERENCE ARTICLES (Accepted/Published)

- [**C16**] Human-AI Collaboration NLP Biomedical Discovery ACL2026 **Yang Wu**, Jinhong Yu, Jingwei Xiong, Zhimin Tao, and Xiaozhong Liu. "Excuse me, may I say something..." CoLabScience, A Proactive AI Assistant for Biomedical Discovery and LLM-Expert Collaborations. In *Proceedings of the 64th Conference of the Association for Computational Linguistics*, 2026
- [**C15**] LLM Trustworthy NLP ACL2026 Findings Rujing Yao, Yufei Shi, **Yang Wu**, Ang Li, Zhuoren Jiang, XiaoFeng Wang, Haixu Tang, and Xiaozhong Liu. Beyond Local vs. External: A Game-Theoretic Framework for Trustworthy Knowledge Acquisition. *Findings of the 64th Conference of the Association for Computational Linguistics*, 2026

- [**C14**] LLM Agents NLP ACL2026 Findings Tong Zhang, **Yang Wu**, Yufei Shi, Rujing Yao, Zhuoren Jiang, and Xiaozhong Liu. From Individual Excellence to Collective Sustainability: Seeking Strategic Equilibrium in Proactive Multi-Agent Teams. *Findings of the 64th Conference of the Association for Computational Linguistics*, 2026
- [**C13**] Chemistry AI LLM NLP ACL2026 Findings Xiaoyi Chen, Mahsa Monshizadeh, Chaoqi Zhang, Jianjun Lang, **Yang Wu**, Genevieve Mortensen, Xiaozhong Liu, and Haixu Tang. LLMs as Lab Engineers: A Benchmark for Analytical Method Lifecycle Management. *Findings of the 64th Conference of the Association for Computational Linguistics*, 2026
- [**C12**] LLM Personalization NLP Information Retrieval EMNLP2026 Yufei Shi, Rujing Yao, Ang Li, **Yang Wu**, Zhuoren Jiang, and Xiaozhong Liu. Not All Memories Are Equal: Hierarchical Collaborative Memory for Validity-Aware Retrieval in LLM Agents. In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*, 2026
- [**C11**] Video Generation NLP EMNLP2026 Workshop REALM **Yang Wu**, Stefano Petrangeli, Ishita Dasgupta, and Yu Shen. CamPilot: A Multi-Agent Cinematic Assistant for Camera-Controlled Video Generation. *REALM Workshop of the 2026 Conference on Empirical Methods in Natural Language Processing*, 2026
- [**C10**] Video Question-Answering LLM NLP EMNLP2026 Workshop NLP4PI Karish Gupta, Matthew Alex, Alex Li, **Yang Wu**, Yun-Wei Chu, Kashif Munir, Xiaotian Zhou, Zhengping Ji, and Xiaozhong Liu. BodyCam-VQA: Enhanced Body-Worn Camera Video Captioning via Multimodal Reasoning and Probe Question Generation. *NLP4PI Workshop of the 2026 Conference on Empirical Methods in Natural Language Processing*, 2026
- [**C9**] Personalized Education NLP AAAI2026 Workshop AI4EDU **Yang Wu**, Rujing Yao, Tong Zhang, Yufei Shi, Zhuoren Jiang, Zhushan Li, and Xiaozhong Liu. Teaching According to Students' Aptitude: Personalized Mathematics Tutoring via Persona-, Memory-, and Forgetting-Aware LLMs. *AI4EDU Workshop of the 40th AAAI Conference on Artificial Intelligence*, 2026
- [**C8**] Legal Intelligence NLP AAAI2026 AI-Law Bridge Rujing Yao, **Yang Wu**, Jinhong Yu, Tong Zhang, Zhuoren Jiang, and Xiaozhong Liu. Who Should Answer? CoT-Guided Two-Stage Routing for Legal QA. *AI-Law Bridge of the 40th AAAI Conference on Artificial Intelligence*, 2026
- [**C7**] Legal Intelligence NLP AAAI2026 AI-Law Bridge Tong Zhang, Yiquan Wu, Wenlin Zhong, Rujing Yao, **Yang Wu**, Yufei Shi, Ang Li, and Xiaozhong Liu. The MCP Quality Paradox: Enhancing Tool Effectiveness in Long-Horizon Reasoning through Phase-Aware State Abstraction. *AI-Law Bridge of the 40th AAAI Conference on Artificial Intelligence*, 2026
- [**C6**] Biomedical Intelligence NLP EMNLP2025 Findings **Yang Wu**, Raha Moraffah, Rujing Yao, Jinhong Yu, Zhimin Tao, and Xiaozhong Liu. Active Domain Knowledge Acquisition with 100-dollar Budget: Enhancing LLMs via Cost-Efficient, Expert-Involved Interaction in Sensitive Domains. *Findings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 2025
- [**C5**] LLM Efficiency NLP EMNLP2025 Findings **Yang Wu**, Huayi Zhang, Yizheng Jiao, Lin Ma, Xiaozhong Liu, Jinhong Yu, Dongyu Zhang, Dezhi Yu, and Wei Xu. Rose: A reward-oriented data selection framework for llm task-specific instruction tuning. *Findings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 2025

- [**C4**] Legal Intelligence NLP NAACL2025 Rujing Yao, **Yang Wu**, Chenghao Wang, Jingwei Xiong, Fang Wang, and Xiaozhong Liu. Elevating Legal LLM Responses: Harnessing Trainable Logical Structures and Semantic Knowledge with Legal Reasoning. In *Proceedings of the 2025 Annual Conference of the North American Chapter of the Association for Computational Linguistics, 2025*
- [**C3**] Legal Intelligence Knowledge Graph WWW2025 Demo Rujing Yao, Yiquan Wu, Tong Zhang, Xuhui Zhang, Yuting Huang, **Yang Wu**, Jiayin Yang, Changlong Sun, Fang Wang, and Xiaozhong Liu. Intelligent Legal Assistant: An Interactive Clarification System for Legal Question Answering. In *Companion Proceedings of the ACM Web Conference 2025 (WWW Companion '25), 2025*
- [**C2**] Legal Intelligence Knowledge Graph ACL2024 Findings **Yang Wu**, Chenghao Wang, Ece Gumusel, and Xiaozhong Liu. Knowledge-Infused Legal Wisdom: Navigating LLM Consultation through the Lens of Diagnostics and Positive-Unlabeled Reinforcement Learning. *Findings of the 62nd Conference of the Association for Computational Linguistics*, 2024
- [**C1**] Chronic Disease Prediction Semi-Supervised Learning CIKM2023 **Yang Wu**, Xurui Li, Xuhong Zhang, Yangyang Kang, Changlong Sun, and Xiaozhong Liu. Community-Based Hierarchical Positive-Unlabeled (PU) Model Fusion for Chronic Disease Prediction. In *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management*, 2023

PEER REVIEWED CONFERENCE ARTICLES (Under Review)

- **[UC4]** Human-AI Collaboration LLM Reasoning NLP ICLR2027 (pending) **Yang Wu**, Kai Zhang, Rujing Yao, Tong Zhang, Yufei Shi, Jinhong Yu, Zhuoren Jiang, and Xiaozhong Liu. Deep Think with Rehearsal for Low-Latency Team-AI Collaboration. *The Fifteenth International Conference on Learning Representations*, 2027
- **[UC3]** Biomedical Intelligence LLM Agents NLP ICLR2027 (pending) Jinhong Yu, Kai Zhang, **Yang Wu**, Roee Shraga, Patricia L. Mabry, Bradley D Gordon, and Xiaozhong Liu. Beyond Patient-Level Triage: Pre-Arrival Care Navigation under Shared, Limited Capacity. *The Fifteenth International Conference on Learning Representations*, 2027
- **[UC2]** LLM Agents Personalized AI NLP AAAI2027 (pending) Yufei Shi, Rujing Yao, Ang Li, **Yang Wu**, Jinhong Yu, Zhuoren Jiang, and Xiaozhong Liu. Not Everyone Needs to Know Everything. *The 41st AAAI Conference on Artificial Intelligence*, 2027
- [**UC1**] LLM Agents Human-AI Collaboration NLP AAAI2027 Demo (Pending) Yufei Shi, Rujing Yao, Tong Zhang, **Yang Wu**, Zhuoren Jiang, and Xiaozhong Liu. TEAM+1: A Proactive, Memory-Grounded AI Teammate for Scientific Collaboration. *Demo of the 41st AAAI Conference on Artificial Intelligence*, 2027

MANUSCRIPTS IN PREPARATION

- [**MP3**] Cybersecurity LLM Agents NLP In Preparation Jinhong Yu, Zhihao Zhang, **Yang Wu**, Tran Ngoc Bao Huynh, Jun Dai, Xiaoyan Sun, XiaoFeng Wang, and Xiaozhong Liu. To Trust, or Not to Trust, That is the Critical Question: Responsible Code Generation with Adversarial Vulnerability Awareness and Calibration. *TBD*
- [**MP2**] Biomedical Intelligence NLP In Preparation Jinhong Yu, Zhikai Xue, **Yang Wu**, Yejin Kim, Patricia L Mabry, Roee Shraga, and Xiaozhong Liu. Bridging Trust and Efficiency: Benchmark Dataset for Collaborative AI-Human Triage for Emergency Room Decision-Making. *TBD*

- [**MP1**] LLM Agents NLP In Preparation Tong Zhang, **Yang Wu**, Yufei Shi, Rujing Yao, Zhuoren Jiang, and Xiaozhong Liu. Proactive Rehearsal: A Learning Framework for Anticipatory Planning in Human-AI Collaboration. *TBD*

PATENT (Pending)

- [**PP1**] Yu Shen, Stefano Petrangeli, Ishita Dasgupta, and **Yang Wu**. Camera-Controlled Video Generation, US Patent Application.

TEACHING

- CS541 / DS541 – Deep Learning, Department of Computer Science, Worcester Polytechnic Institute, Jan 2025 – Apr 2025 (Teaching Assistant)
- CS553 / DS553 – Machine Learning Development and Operations, Department of Computer Science, Worcester Polytechnic Institute, Aug 2024 – Dec 2024 (Teaching Assistant)
- CS2223 – Algorithms, Department of Computer Science, Worcester Polytechnic Institute, Mar 2024 – May 2024 (Teaching Assistant)
- CS1101 – Introduction to Program Design, Department of Computer Science, Worcester Polytechnic Institute, Jan 2024 – Mar 2024 (Teaching Assistant)
- CS4342 – Machine Learning, Department of Computer Science, Worcester Polytechnic Institute, Oct 2023 – Dec 2023 (Teaching Assistant)
- CS2223 – Algorithms, Department of Computer Science, Worcester Polytechnic Institute, Aug 2023 – Oct 2023 (Teaching Assistant)
- CS2223 – Algorithms, Department of Computer Science, Worcester Polytechnic Institute, Mar 2023 – May 2023 (Teaching Assistant)
- CS3133 – Foundations of Computer Science, Department of Computer Science, Worcester Polytechnic Institute, Oct 2022 – Dec 2022 (Teaching Assistant)
- CS1101 – Introduction to Program Design, Department of Computer Science, Worcester Polytechnic Institute, Aug 2022 – Oct 2022 (Teaching Assistant)
- CS2119 – Application Building with Object-Oriented Concepts, Department of Computer Science, Worcester Polytechnic Institute, Mar 2022 – May 2022 (Teaching Assistant)
- CS4341 – Introduction to Artificial Intelligence, Department of Computer Science, Worcester Polytechnic Institute, Jan 2022 – Mar 2022 (Teaching Assistant)
- CS4342 – Machine Learning, Department of Computer Science, Worcester Polytechnic Institute, Oct 2021 – Dec 2021 (Teaching Assistant)
- CS1101 – Introduction to Program Design, Department of Computer Science, Worcester Polytechnic Institute, Aug 2021 – Oct 2021 (Teaching Assistant)

AWARDS AND ACHIEVEMENTS

- **AAAI 2026 AI-Law Bridge Best Paper Award (2026)** – The 40th AAAI Conference on Artificial Intelligence Bridge between Artificial Intelligence and Law
- **Rank No. 2 (2023)** for Food for Thought Challenge Competition, United States Department of Agriculture (USDA)

REVIEW AND COMMITTEE

- Conference Reviewer: The 41st AAAI Conference on Artificial Intelligence (AAAI2027) *(4 papers reviewed)*

- Conference Reviewer: The 40th Annual Conference on Neural Information Processing Systems (NeurIPS2026) *(4 papers Reviewed)*
- Conference Reviewer: The 40th AAAI Conference on Artificial Intelligence (AAAI2026) *(4 papers reviewed)*
- Conference Reviewer: The 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP2026) *(4 papers Reviewed)*
- Conference Reviewer: The 64th Annual Meeting of the Association for Computational Linguistics (ACL2026) *(8 papers reviewed)*
- Conference Reviewer: The 30th Conference on Computational Natural Language Learning (CoNLL2026) *(2 papers Reviewed)*
- Conference Reviewer: The 63rd Annual Meeting of the Association for Computational Linguistics (ACL2025) *(16 papers reviewed)*
- Conference Reviewer: The 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP2025) *(4 papers reviewed)*
- Conference Reviewer: The 18th Conference of the European Chapter of the Association for Computational Linguistics (EACL2025) *(2 papers reviewed)*
- Conference Reviewer: The 33rd ACM International Conference on Information and Knowledge Management (CIKM2024) *(2 papers reviewed)*

SELECTED PROJECT

Team+1: Proactive LLM Assistants for Scientific Discovery **Worcester, MA**

Project Lead *2025 - Present*

- Scientific discovery depends on timely input, but most AI assistants only react when asked, and real-time reasoning is often too slow to help in the moment. This project is to build a proactive LLM assistant that anticipates when to intervene and pre-computes reasoning offline to support low-latency biomedical collaboration.

Data-Efficient LLM Training **Worcester, MA**

Project Lead *2024 - Present*

- Training large language models at full scale is costly, and much of the data used rarely improves performance. This project is to develop a reward-oriented, active-learning-driven framework that selects only the most valuable training data for LLM instruction tuning.

Human-Centered and Multimodal LLM Assistants **Worcester, MA**

Project Lead *2023 - Present*

- Legal reasoning, personalized learning, and creative production each require AI that understands domain knowledge and individual user needs, not just language. This project is to design human-centered, multimodal LLM assistants spanning legal reasoning, personalized tutoring, and camera-controlled movie generation.

PEER REVIEWED CONFERENCE ORAL PRESENTATION

- **Wu, Y.,** Yu, J., Xiong, J., Tao, Z., &Liu, X., (2026) "Excuse me, may I say something..." CoLabScience, A Proactive AI Assistant for Biomedical Discovery and LLM-Expert Collaborations, Proceedings of the 64th Conference of the Association for Computational Linguistics. ACL2026.
- **Wu, Y.,** Li, X., Zhang, X., Kang, Y., Sun, C., &Liu, X., (2023) Community-Based Hierarchical Positive-Unlabeled (PU) Model Fusion for Chronic Disease Prediction, Proceedings of the 32nd ACM International Conference on Information and Knowledge Management. CIKM2023.